

Unconscious Visual Effects for Enhancing Concentration in Virtual Reality

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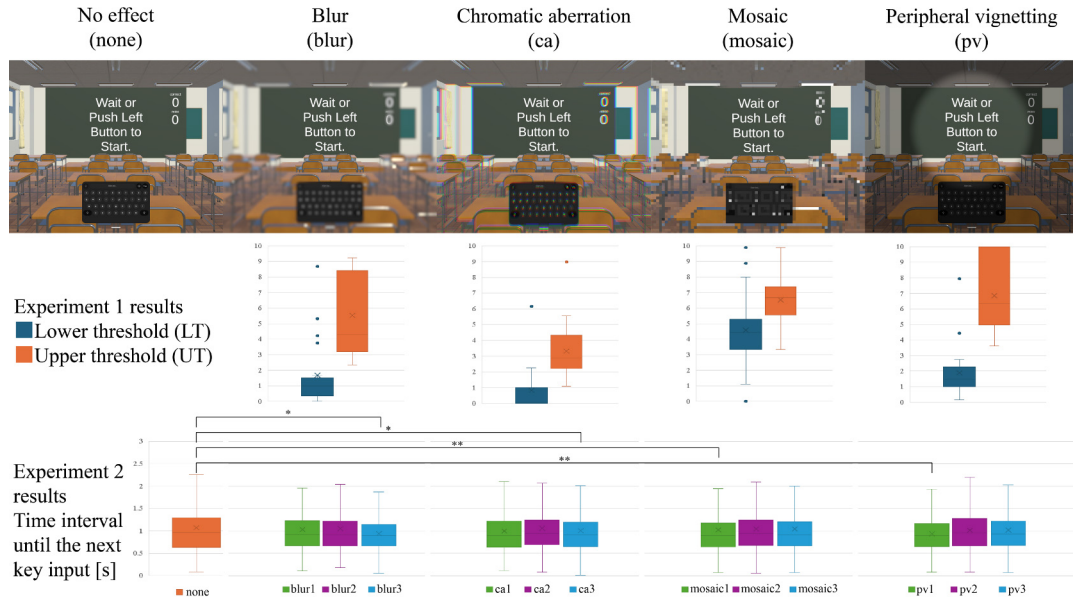


Figure 1: The view when the effects used in each experiment were applied, and the results of Experiments 1 and 2. The name (abbreviation) of each effect is listed on the top row. The next row shows the screenshot of each effect. In Experiment 1, the effect amount noticed for each effect was determined using the limit method. The box plots show the responses of each experimental participant when the upper and lower discrimination thresholds were investigated. In Experiment 2, we used a typing task to experiment with three levels of effect amounts within the range obtained in Experiment 1 for each effect, to see if concentration increased. This figure shows the interval between key inputs, which was found to be particularly significant in the data collected. The graphs are marked with * for pairs with p -values below the 5% significance level, ** for pairs with p -values below the 1% significance level, respectively.

ABSTRACT

This study investigated the effectiveness of unconscious visual effects in VR for enhancing concentration. The research evaluated four visual effects (blur, chromatic aberration, mosaic, peripheral vignetting) applied to peripheral vision through two experiments: first determining detection thresholds, then assessing concentration impact through typing tasks. 20 participants identified effect thresholds using the method of limits, followed by 15 participants testing concentration effects. Results showed blur and peripheral vignetting significantly improved typing accuracy and reduced keystroke intervals, supported by subjective evaluations. The research demonstrates potential for enhancing concentration through peripheral visual modifications without conscious awareness.

Index Terms: Concentration, visual effects, peripheral Vision

1 INTRODUCTION

While VR/AR systems are often evaluated in terms of usability, efficiency, and novel experiential value, demonstrating the advantages of wearable devices requires showing effective benefits that arise

simply from wearing them compared to not wearing them. We hypothesized that applying subtle effects to users' peripheral vision while wearing an HMD, without their conscious awareness, could enhance concentration. Hillaire et al. proposed a method to improve camera motion and depth-of-field blur effects in VR using an eye-tracking system [2]. Their research achieved a more natural visual experience by adjusting camera movements based on the user's gaze point. Yamaura et al. proposed a method to enhance digital content viewing experience by applying Gaussian blur effects to peripheral vision [3]. Buhler et al. proposed two methods applying visual effects to peripheral vision to reduce VR sickness [1]. These studies demonstrate that properly controlled visual effects can improve the quality of VR experiences.

This study proposes a method to enhance concentration while minimizing psychological burden by applying subtle visual effects to peripheral vision below the threshold of conscious awareness. Our method employs 4 types of visual effects: blur, chromatic aberration, mosaic and peripheral vignetting.

2 PROPOSED METHOD

We propose a method to enhance concentration by applying unconscious visual effects to the peripheral vision of HMD users. Using psychophysical techniques, we experimentally determine the effect size of visual effects acting unconsciously. Furthermore, in a second stage of the experiment, we will investigate which and how much effect size of which visual effect is suitable for improving concentration. In Experiment 1, we measured visual effect detec-

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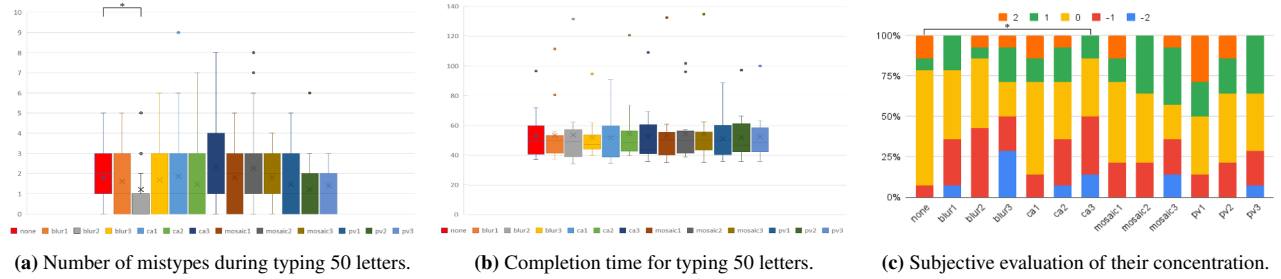


Figure 2: Results of Experiment 2. The graphs are marked with * for pairs with p -values below the 5% significance level.

tion thresholds based on the method of limits, i. e., identifying the unnoticeable effect amounts for four effects (blur, chromatic aberration, mosaic, and peripheral vignetting) through ascending and descending series. The system was implemented using Unity, applied to a range of viewing angles from 70 to 120 degrees, and implemented so that the amount of effect could be adjusted so that it could be measured in the experiment.

In Experiment 2, we evaluate how these visual effects at the thresholds determined in Experiment 1 impact concentration. Quantitatively measure the effectiveness of each visual effect using a typing task on VR. A system was implemented to type 50 randomly arranged letters of the alphabet by key selection on a VR keyboard in sequence. The system records the number of misspelled letters, the time it takes to finish typing, and the interval between keystrokes. The sequence of the alphabet is adjusted so that the same letter is not followed in succession. Participants' concentration levels are comprehensively evaluated using task performance metrics and subjective assessments.

3 EXPERIMENTS

The procedure for each experiment is described below.

Experiment 1: Threshold Measurement

We measured recognition thresholds for each visual effect with 20 participants (aged 19-45 years). Using the method of limits, we measured thresholds in both ascending and descending series. The experimental procedure was as follows:

- I. Participants wore the HMD and performed a typing task.
- II. The intensity of visual effects was gradually changed during the task, and participants reported when they noticed changes.
- III. For each visual effect, measurements were taken in both ascending and descending series.
- IV. Recognition thresholds were determined from the upper and lower differential thresholds.

Experiment 2: Concentration Effect Evaluation

We evaluated concentration effects with thresholds from Experiment 1. 15 participants (aged 19-26 years) followed this procedure:

- I. Participants performed random English letter typing task.
- II. Measurements were taken for each of the five visual effects, with and without the effects applied.
- III. Performance metrics recorded included typing errors, character count, and subjective concentration levels.

The order of effects was randomized to eliminate order bias. Additionally, a 30-second to 1-minute breaks and subjective feedback periods were provided between trials.

4 RESULTS

The results of Experiment 1 are shown in Figure 1. From these results, the lower and upper discrimination thresholds and their averages are obtained, numbered from 1 to 3, starting from the smallest to the largest, and are shown together with the abbreviations in the graph of results. (e.g., "ca1" indicates that the result uses the lower discrimination threshold value for chromatic aberration).

A box plot diagram of the number of typing errors and the time taken to type when each effect is applied in Experiment 2 is shown in Figure 2 (a) and (b). The interval time for key typing is shown in on the bottom line of Figure 1. Since these objective measures follow a normal distribution, we took a t -test at a significance level of 5% between each effect to check for significant differences. There was no significant difference in total time between effects, but the mean time was shorter with effects than without. We considered work accuracy as a factor to measure concentration and measured the number of mistyping. "Blur2" significantly reduced mistyping. Other effects also showed a decreasing trend in mistyping. Keystroke intervals were significantly different for the four effects compared to no effects. In objective measures, the application of effects to peripheral vision was shown to increase concentration.

Figure 2 (c) shows subjective evaluation results. The participants rated their concentration ability on a 5-point scale from "-2: I could not concentrate" to "2: I could concentrate". A Wilcoxon signed rank test was used to test the significance level of 5%. The results showed that "dark1" was the most concentrated. This subjective evaluation was consistent with Time Interval scores. "Ca3" was reported to be significantly less concentrated than "none".

5 CONCLUSION AND FUTURE WORK

This study investigated unconscious visual effects in VR to enhance concentration. Through two experiments, we evaluated 4 different visual effects applied to users' peripheral vision. Experiment 1 determined detection thresholds for each effect, while Experiment 2 assessed their impact on concentration through a typing task. Results showed that blur and peripheral vignetting significantly improved typing accuracy and reduced keystroke intervals, with subjective evaluations supporting these findings.

Future issues include investigating individual differences by analyzing effects in various user groups, and finding a more focused state by collecting and analyzing physiological characteristics (EEG, eye tracking, heart rate data) in an integrated manner.

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REFERENCES

- [1] H. Buhler, S. Misztal, and J. Schild. Reducing vr sickness through peripheral visual effects. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 517–9, 2018. doi: 10.1109/VR.2018.8446346 1
- [2] S. Hillaire, A. Lecuyer, R. Cozot, and G. Casiez. Using an eye-tracking system to improve camera motions and depth-of-field blur effects in virtual environments. In *2008 IEEE Virtual Reality Conference*, pp. 47–50, 2008. doi: 10.1109/VR.2008.4480749 1
- [3] H. Yamaura, M. Tamura, and S. Nakamura. Image blurring method for enhancing digital content viewing experience. In M. Kurosu, ed., *Human-Computer Interaction. Theories, Methods, and Human Issues*, pp. 355–370. Springer International Publishing, Cham, 2018. 1